COURSE OUTLINE

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# Agentic Virtual Patching Platform

From vulnerability finding to tested interim mitigation

3 days (8 hours per day) | Intermediate to advanced | Instructor-led, hands-on labs | Capstone project

## About this course

### Who should attend

Application security, product security, DevSecOps, WAF and platform engineers, vulnerability management teams.

### Prerequisites

- Basic HTTP and web application security
- Reverse proxy and WAF concepts
- Python and container basics
- Familiarity with an AI coding assistant is helpful, not required

### What you will be able to do

- Explain the role and limits of virtual patching as an interim mitigation, not a substitute for a permanent fix
- Build a finding intake and evidence and context model with provenance
- Generate candidate rules for a lab enforcement point using constrained templates and strict schemas
- Create positive, negative, regression and bypass-oriented test cases and fail closed on incomplete validation
- Produce an approval package with test evidence, rollback instructions and monitoring criteria

## Tools and technologies

Claude | Python | FastAPI | Pydantic | pytest | Docker Compose | NGINX / ModSecurity lab | OWASP | SQLite

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GitHub

## Table of contents

**DAY 1** 3 modules

MODULE 1

### Virtual patching fundamentals and the agentic baseline

- What a virtual patch can and cannot do; interim mitigation versus permanent fix
- Threat model, HTTP and WAF basics, the lab enforcement point and deliberately vulnerable endpoint
- Shared engineering principles: model output as a proposal, authorization outside the model, human approval
- The AI-assisted development workflow used in every build

**HANDS-ON LAB**
Stand up the lab with Docker Compose and exercise the vulnerable endpoint behind the proxy.

**DELIVERABLE**
Reproducible lab and product brief.

MODULE 2

### Finding ingestion and evidence model

- Typed vulnerability record: advisory ID, affected component and endpoint, evidence references, confidence, source timestamp
- Mock scanner JSON importer with schema validation and provenance
- Invalid-input cases and rejected-row reporting

**HANDS-ON LAB**
Implement the finding normalizer and unit tests with the AI coding assistant; inspect every diff.

**DELIVERABLE**
Working finding intake.

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MODULE 3

## Context collection

- Endpoint, application owner, environment and route metadata from fixtures or the demo app
- Never inferring reachability without evidence
- Marking unknowns explicitly

**HANDS-ON LAB**
Build the context collector and attach it to each finding.

**DELIVERABLE**
Finding plus context bundle.

**DAY 2** 3 modules

MODULE 4

## Agentic mitigation planning

- An agent summarizes the issue and proposes a strategy from an approved playbook
- Citations to supplied evidence are mandatory; unknowns are marked
- Structured outputs, evidence-grounded reasoning, model adapter layer

**HANDS-ON LAB**
Write the planning prompt and output schema; test it with missing and contradictory evidence.

**DELIVERABLE**
Mitigation planning agent.

MODULE 5

## Constrained candidate rule generation

- Rules generated only through approved, versioned templates and a schema
- Rejecting arbitrary directives and anything outside the lab's supported format
- Rule selection logic and its tests

**HANDS-ON LAB**
Generate candidate rules for the lab finding and prove the generator refuses out-of-scope rules.

**DELIVERABLE**
Template-constrained rule generator.

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MODULE 6

## Test harness and validation gate

- Positive tests for the target case, benign traffic tests, regression tests, bypass variations
- Running against the isolated proxy or WAF and scoring blocked, allowed and limitations
- Fail closed when validation is incomplete; confidence reporting

**HANDS-ON LAB**
Build the harness, run it, and break a rule with a bypass variant to see the gate fail closed.

**DELIVERABLE**
Automated validation report.

**DAY 3** 3 modules

MODULE 7

## Approval package and staging deployment

- Review bundle: diff, test results, owner, expiry, rollback command, approver
- Deploy only to lab staging after explicit approval; the deployment adapter
- Expiry and re-validation of interim mitigations

**HANDS-ON LAB**
Produce the approval package and deploy to staging through the gated adapter.

**DELIVERABLE**
Approval and deployment workflow.

MODULE 8

## Monitoring, rollback and operations

- Request and block metrics from synthetic traffic
- Rollback exercise and operator runbook
- Audit events, versioned artifacts, known limitations

**HANDS-ON LAB**
Run the rollback drill and write the runbook.

**DELIVERABLE**
Monitoring and rollback runbook.

MODULE 9

## Capstone: end-to-end virtual patching lab platform

- Finding in, validated mitigation out, with no deployment without approval
- Acceptance criteria review, safety and limitations statement, demo

**HANDS-ON LAB**
Capstone build and demo.

**DELIVERABLE**
End-to-end platform and runbook.

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## Capstone

### Capstone project

A repeatable lab platform that ingests a finding, proposes an evidence-linked mitigation, validates it against target and benign traffic, and ships an approval package with tested rollback.

## Assessment

| Assessment area | Weight | Evidence |
|---|---|---|
| Working functionality | 25% | Demonstrable end-to-end workflow in a repeatable lab |
| Security and safety controls | 20% | Threat model, authorization boundaries, approval gates, safe failure |
| Testing and evaluation | 20% | Automated tests, golden dataset, measured quality and known limitations |
| Engineering quality | 15% | Clear interfaces, error handling, reproducible setup, maintainable code |
| Evidence and explainability | 10% | Traceable findings, sources, assumptions, logs |
| Documentation and presentation | 10% | Architecture diagram, README, runbook, demo, lessons learned |

## Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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